POKHARA UNIVERSITY

Level: Bachelor
Programme:B E
Course:Computer Architecture

Semester:Fall

Year : 2020
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Differentiate Computer Organization and architecture. Explain the various addressing modes. 8
   b) Define RTL. Explain about BUS and Memory transferin RTL. 7
2. a) Explain different types of registers used in computer system. What is use of register in computer system? 7
   b) Perform multiplication -13*7 using Booth's Algorithm. 8
3. a) What are the various number representation method in computer system? 7
   b) Explain block diagram of micro programmed control organization and write down its advantages. 8
4. a) Explain associative and set-associative mapping in cache memory? 8
   b) Define an interrupt with its classes in detail. Also explain about interrupt driven I/O. 7
5. a) What is pipelining hazards? Explain structural and data hazard and its solution 7
   b) What is cache coherence problem? Explain software and hardware solution of cache coherence problem. 8
6. a) Write about parallelism in computer system. Explain about symmetric multiprocessor in detail. 7
   b) Explain about the various Hardware Performance issues in Multicore Computers. 8

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7. Write short notes on: (Any two)

a) Shift Micro operations

b) Magnetic tape and disk

c) Differentiate between RISC and CISC.

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